Lesson Planning

Complete Lesson Plan Book

Class: Class 11 – ICS

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Introduction to Computer and IT

Unit 2: Computer System Components

Unit 3: Data Representation

Unit 4: System Software and Operating Systems

Unit 5: Application Software

Unit 6: Introduction to Programming

Unit 7: C Language Basics

Unit 8: Data Types, Operators and I/O

Unit 9: Control Structures in C

Unit 10: Functions and Arrays in C

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Unit 1: Introduction to Computer and IT

Learning Objectives

• Define and explain computer generations in the context of Introduction to Computer and IT.

• Differentiate between types of computers and information technology with suitable examples.

• Solve conceptual problems related to uses of computers accurately.

• Apply the ideas of Introduction to Computer and IT to everyday situations involving computer ethics.

Key Concepts

• Meaning, definition and scope of computer generations.

• Important properties, types and examples of types of computers.

• Relationship between information technology and uses of computers.

• Practical applications of Introduction to Computer and IT, especially computer ethics.

• Key terms and a summary of the main points of Introduction to Computer and IT.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Introduction to Computer and IT with a daily-life example.
2. Explain computer generations on the board with definitions and clear diagrams.
3. Discuss types of computers and information technology using simple, relatable examples.
4. Involve the students through short questions on uses of computers.
5. Solve an exercise question on computer ethics together with the class.
6. Summarize the lesson and highlight the key points of Introduction to Computer and IT.

Classroom Activities

• Group discussion: students compare types of computers with information technology.

• Board activity: solve a question on uses of computers in pairs.

• Quick quiz on the key terms of Introduction to Computer and IT.

Assessment Questions

1. Define computer generations.
2. Differentiate between types of computers and information technology.
3. Explain with an example: uses of computers.
4. Write a short note on computer ethics.

Homework

• Read Introduction to Computer and IT from the book and solve the exercise questions.

• Prepare short notes on computer generations and types of computers.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Computer System Components

Learning Objectives

• Define and explain input devices in the context of Computer System Components.

• Differentiate between output devices and CPU with suitable examples.

• Solve conceptual problems related to memory types accurately.

• Apply the ideas of Computer System Components to everyday situations involving storage devices.

Key Concepts

• Meaning, definition and scope of input devices.

• Important properties, types and examples of output devices.

• Relationship between CPU and memory types.

• Practical applications of Computer System Components, especially storage devices.

• Key terms and a summary of the main points of Computer System Components.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Computer System Components with a daily-life example.
2. Explain input devices on the board with definitions and clear diagrams.
3. Discuss output devices and CPU using simple, relatable examples.
4. Involve the students through short questions on memory types.
5. Solve an exercise question on storage devices together with the class.
6. Summarize the lesson and highlight the key points of Computer System Components.

Classroom Activities

• Group discussion: students compare output devices with CPU.

• Board activity: solve a question on memory types in pairs.

• Quick quiz on the key terms of Computer System Components.

Assessment Questions

1. Define input devices.
2. Differentiate between output devices and CPU.
3. Explain with an example: memory types.
4. Write a short note on storage devices.

Homework

• Read Computer System Components from the book and solve the exercise questions.

• Prepare short notes on input devices and output devices.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Data Representation

Learning Objectives

• Define and explain number systems in the context of Data Representation.
• Differentiate between binary conversion and ASCII code with suitable examples.
• Solve conceptual problems related to Boolean logic accurately.
• Apply the ideas of Data Representation to everyday situations involving logic gates.

Key Concepts

• Meaning, definition and scope of number systems.
• Important properties, types and examples of binary conversion.
• Relationship between ASCII code and Boolean logic.
• Practical applications of Data Representation, especially logic gates.
• Key terms and a summary of the main points of Data Representation.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Data Representation with a daily-life example.
2. Explain number systems on the board with definitions and clear diagrams.
3. Discuss binary conversion and ASCII code using simple, relatable examples.
4. Involve the students through short questions on Boolean logic.
5. Solve an exercise question on logic gates together with the class.
6. Summarize the lesson and highlight the key points of Data Representation.

Classroom Activities

• Group discussion: students compare binary conversion with ASCII code.
• Board activity: solve a question on Boolean logic in pairs.
• Quick quiz on the key terms of Data Representation.

Assessment Questions

1. Define number systems.
2. Differentiate between binary conversion and ASCII code.
3. Explain with an example: Boolean logic.
4. Write a short note on logic gates.

Homework

• Read Data Representation from the book and solve the exercise questions.
• Prepare short notes on number systems and binary conversion.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: System Software and Operating Systems

Learning Objectives

• Define and explain operating system functions in the context of System Software and Operating Systems.

• Differentiate between types of OS and utility programs with suitable examples.

• Solve conceptual problems related to device drivers accurately.

• Apply the ideas of System Software and Operating Systems to everyday situations involving file management.

Key Concepts

• Meaning, definition and scope of operating system functions.

• Important properties, types and examples of types of OS.

• Relationship between utility programs and device drivers.

• Practical applications of System Software and Operating Systems, especially file management.

• Key terms and a summary of the main points of System Software and Operating Systems.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce System Software and Operating Systems with a daily-life example.
2. Explain operating system functions on the board with definitions and clear diagrams.
3. Discuss types of OS and utility programs using simple, relatable examples.
4. Involve the students through short questions on device drivers.
5. Solve an exercise question on file management together with the class.
6. Summarize the lesson and highlight the key points of System Software and Operating Systems.

Classroom Activities

• Group discussion: students compare types of OS with utility programs.

• Board activity: solve a question on device drivers in pairs.

• Quick quiz on the key terms of System Software and Operating Systems.

Assessment Questions

1. Define operating system functions.
2. Differentiate between types of OS and utility programs.
3. Explain with an example: device drivers.
4. Write a short note on file management.

Homework

• Read System Software and Operating Systems from the book and solve the exercise questions.

• Prepare short notes on operating system functions and types of OS.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Application Software

Learning Objectives

• Define and explain word processors in the context of Application Software.
• Differentiate between spreadsheets and presentation software with suitable examples.
• Solve conceptual problems related to database software accurately.
• Apply the ideas of Application Software to everyday situations involving software suites.

Key Concepts

• Meaning, definition and scope of word processors.
• Important properties, types and examples of spreadsheets.
• Relationship between presentation software and database software.
• Practical applications of Application Software, especially software suites.
• Key terms and a summary of the main points of Application Software.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Application Software with a daily-life example.
2. Explain word processors on the board with definitions and clear diagrams.
3. Discuss spreadsheets and presentation software using simple, relatable examples.
4. Involve the students through short questions on database software.
5. Solve an exercise question on software suites together with the class.
6. Summarize the lesson and highlight the key points of Application Software.

Classroom Activities

• Group discussion: students compare spreadsheets with presentation software.
• Board activity: solve a question on database software in pairs.
• Quick quiz on the key terms of Application Software.

Assessment Questions

1. Define word processors.
2. Differentiate between spreadsheets and presentation software.
3. Explain with an example: database software.
4. Write a short note on software suites.

Homework

• Read Application Software from the book and solve the exercise questions.
• Prepare short notes on word processors and spreadsheets.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Introduction to Programming

Learning Objectives

• Define and explain algorithms in the context of Introduction to Programming.

• Differentiate between flowcharts and pseudocode with suitable examples.

• Solve conceptual problems related to programming languages accurately.

• Apply the ideas of Introduction to Programming to everyday situations involving language translators.

Key Concepts

• Meaning, definition and scope of algorithms.

• Important properties, types and examples of flowcharts.

• Relationship between pseudocode and programming languages.

• Practical applications of Introduction to Programming, especially language translators.

• Key terms and a summary of the main points of Introduction to Programming.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Introduction to Programming with a daily-life example.
2. Explain algorithms on the board with definitions and clear diagrams.
3. Discuss flowcharts and pseudocode using simple, relatable examples.
4. Involve the students through short questions on programming languages.
5. Solve an exercise question on language translators together with the class.
6. Summarize the lesson and highlight the key points of Introduction to Programming.

Classroom Activities

• Group discussion: students compare flowcharts with pseudocode.

• Board activity: solve a question on programming languages in pairs.

• Quick quiz on the key terms of Introduction to Programming.

Assessment Questions

1. Define algorithms.
2. Differentiate between flowcharts and pseudocode.
3. Explain with an example: programming languages.
4. Write a short note on language translators.

Homework

• Read Introduction to Programming from the book and solve the exercise questions.

• Prepare short notes on algorithms and flowcharts.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: C Language Basics

Learning Objectives

• Define and explain history of C in the context of C Language Basics.
• Differentiate between program structure and header files with suitable examples.
• Solve conceptual problems related to main function accurately.
• Apply the ideas of C Language Basics to everyday situations involving comments.

Key Concepts

• Meaning, definition and scope of history of C.
• Important properties, types and examples of program structure.
• Relationship between header files and main function.
• Practical applications of C Language Basics, especially comments.
• Key terms and a summary of the main points of C Language Basics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce C Language Basics with a daily-life example.
2. Explain history of C on the board with definitions and clear diagrams.
3. Discuss program structure and header files using simple, relatable examples.
4. Involve the students through short questions on main function.
5. Solve an exercise question on comments together with the class.
6. Summarize the lesson and highlight the key points of C Language Basics.

Classroom Activities

• Group discussion: students compare program structure with header files.
• Board activity: solve a question on main function in pairs.
• Quick quiz on the key terms of C Language Basics.

Assessment Questions

1. Define history of C.
2. Differentiate between program structure and header files.
3. Explain with an example: main function.
4. Write a short note on comments.

Homework

• Read C Language Basics from the book and solve the exercise questions.
• Prepare short notes on history of C and program structure.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Data Types, Operators and I/O

Learning Objectives

• Define and explain data types in the context of Data Types, Operators and I/O.

• Differentiate between variables and operators with suitable examples.

• Solve conceptual problems related to printf and scanf accurately.

• Apply the ideas of Data Types, Operators and I/O to everyday situations involving type conversion.

Key Concepts

• Meaning, definition and scope of data types.

• Important properties, types and examples of variables.

• Relationship between operators and printf and scanf.

• Practical applications of Data Types, Operators and I/O, especially type conversion.

• Key terms and a summary of the main points of Data Types, Operators and I/O.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Data Types, Operators and I/O with a daily-life example.
2. Explain data types on the board with definitions and clear diagrams.
3. Discuss variables and operators using simple, relatable examples.
4. Involve the students through short questions on printf and scanf.
5. Solve an exercise question on type conversion together with the class.
6. Summarize the lesson and highlight the key points of Data Types, Operators and I/O.

Classroom Activities

• Group discussion: students compare variables with operators.

• Board activity: solve a question on printf and scanf in pairs.

• Quick quiz on the key terms of Data Types, Operators and I/O.

Assessment Questions

1. Define data types.
2. Differentiate between variables and operators.
3. Explain with an example: printf and scanf.
4. Write a short note on type conversion.

Homework

• Read Data Types, Operators and I/O from the book and solve the exercise questions.

• Prepare short notes on data types and variables.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Control Structures in C

Learning Objectives

• Define and explain if statement in the context of Control Structures in C.

• Differentiate between switch statement and for loop with suitable examples.

• Solve conceptual problems related to while loop accurately.

• Apply the ideas of Control Structures in C to everyday situations involving nested loops.

Key Concepts

• Meaning, definition and scope of if statement.

• Important properties, types and examples of switch statement.

• Relationship between for loop and while loop.

• Practical applications of Control Structures in C, especially nested loops.

• Key terms and a summary of the main points of Control Structures in C.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Control Structures in C with a daily-life example.
2. Explain if statement on the board with definitions and clear diagrams.
3. Discuss switch statement and for loop using simple, relatable examples.
4. Involve the students through short questions on while loop.
5. Solve an exercise question on nested loops together with the class.
6. Summarize the lesson and highlight the key points of Control Structures in C.

Classroom Activities

• Group discussion: students compare switch statement with for loop.

• Board activity: solve a question on while loop in pairs.

• Quick quiz on the key terms of Control Structures in C.

Assessment Questions

1. Define if statement.
2. Differentiate between switch statement and for loop.
3. Explain with an example: while loop.
4. Write a short note on nested loops.

Homework

• Read Control Structures in C from the book and solve the exercise questions.

• Prepare short notes on if statement and switch statement.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: Functions and Arrays in C

Learning Objectives

• Define and explain function definition in the context of Functions and Arrays in C.
• Differentiate between function call and arrays with suitable examples.
• Solve conceptual problems related to strings accurately.
• Apply the ideas of Functions and Arrays in C to everyday situations involving recursion basics.

Key Concepts

• Meaning, definition and scope of function definition.
• Important properties, types and examples of function call.
• Relationship between arrays and strings.
• Practical applications of Functions and Arrays in C, especially recursion basics.
• Key terms and a summary of the main points of Functions and Arrays in C.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Functions and Arrays in C with a daily-life example.
2. Explain function definition on the board with definitions and clear diagrams.
3. Discuss function call and arrays using simple, relatable examples.
4. Involve the students through short questions on strings.
5. Solve an exercise question on recursion basics together with the class.
6. Summarize the lesson and highlight the key points of Functions and Arrays in C.

Classroom Activities

• Group discussion: students compare function call with arrays.
• Board activity: solve a question on strings in pairs.
• Quick quiz on the key terms of Functions and Arrays in C.

Assessment Questions

1. Define function definition.
2. Differentiate between function call and arrays.
3. Explain with an example: strings.
4. Write a short note on recursion basics.

Homework

• Read Functions and Arrays in C from the book and solve the exercise questions.
• Prepare short notes on function definition and function call.

Time Allocation

4–6 periods of 40 minutes each